Task – 5

Wireshark (Packet capturing and filtering)

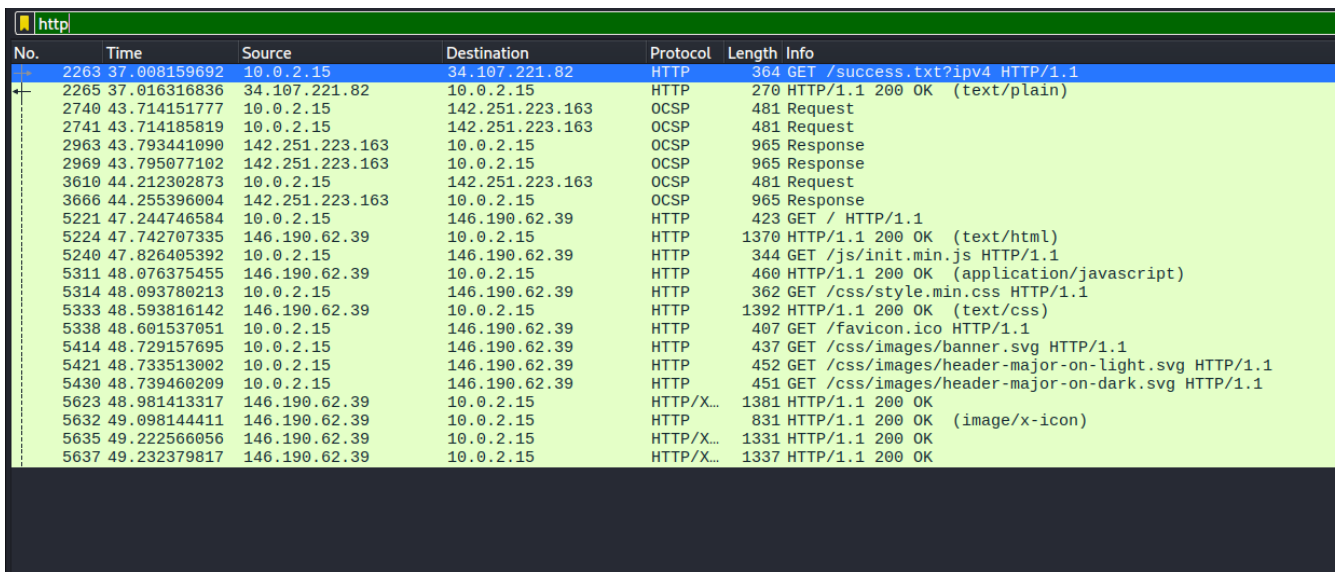
Step 1: Install the wireshark and start the capture.

Step 2: Keep the capturing turned on for a minute atleast, meanwhile, in order to create network traffic, open a few websites or ping any known domains.

Step 3: Stop the capture, and start filtering the packets based on the protocols.

Filtering what is specified in the task sheet,

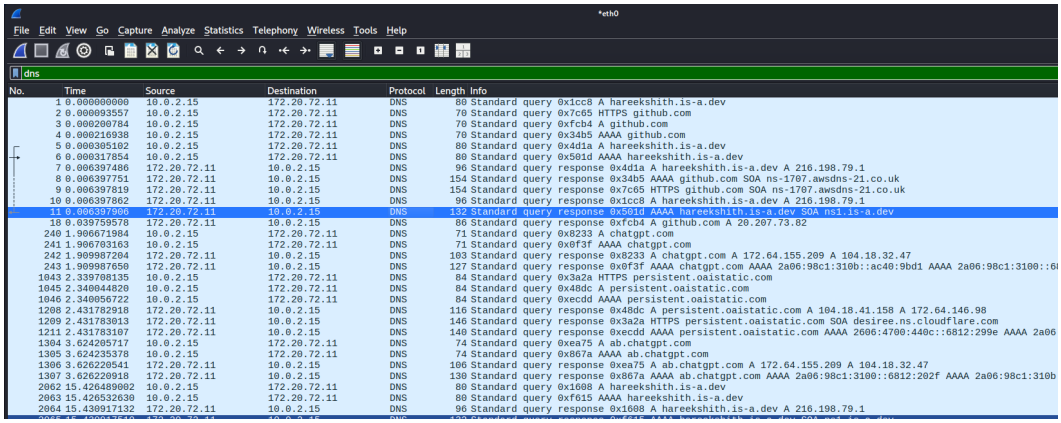
HTTP



http

No.	Time	Source	Destination	Protocol	Length	Info
2263	37.008159692	10.0.2.15	34.107.221.82	HTTP	364	GET /success.txt?ipv4 HTTP/1.1
2265	37.016316836	34.107.221.82	10.0.2.15	HTTP	270	HTTP/1.1 200 OK (text/plain)
2740	43.714151777	10.0.2.15	142.251.223.163	OCSP	481	Request
2741	43.714185819	10.0.2.15	142.251.223.163	OCSP	481	Request
2963	43.793441090	142.251.223.163	10.0.2.15	OCSP	965	Response
2969	43.795077102	142.251.223.163	10.0.2.15	OCSP	965	Response
3610	44.212302873	10.0.2.15	142.251.223.163	OCSP	481	Request
3666	44.255396004	142.251.223.163	10.0.2.15	OCSP	965	Response
5221	47.244746584	10.0.2.15	146.190.62.39	HTTP	423	GET / HTTP/1.1
5224	47.742707335	146.190.62.39	10.0.2.15	HTTP	1370	HTTP/1.1 200 OK (text/html)
5240	47.826405392	10.0.2.15	146.190.62.39	HTTP	344	GET /js/init.min.js HTTP/1.1
5311	48.076375455	146.190.62.39	10.0.2.15	HTTP	460	HTTP/1.1 200 OK (application/javascript)
5314	48.093780213	10.0.2.15	146.190.62.39	HTTP	362	GET /css/style.min.css HTTP/1.1
5333	48.593816142	146.190.62.39	10.0.2.15	HTTP	1392	HTTP/1.1 200 OK (text/css)
5338	48.601537051	10.0.2.15	146.190.62.39	HTTP	407	GET /favicon.ico HTTP/1.1
5414	48.729157695	10.0.2.15	146.190.62.39	HTTP	437	GET /css/images/banner.svg HTTP/1.1
5421	48.733513002	10.0.2.15	146.190.62.39	HTTP	452	GET /css/images/header-major-on-light.svg HTTP/1.1
5430	48.739460209	10.0.2.15	146.190.62.39	HTTP	451	GET /css/images/header-major-on-dark.svg HTTP/1.1
5623	48.981413317	146.190.62.39	10.0.2.15	HTTP/X…	1381	HTTP/1.1 200 OK
5632	49.098144411	146.190.62.39	10.0.2.15	HTTP	831	HTTP/1.1 200 OK (image/x-icon)
5635	49.222566056	146.190.62.39	10.0.2.15	HTTP/X…	1331	HTTP/1.1 200 OK
5637	49.232379817	146.190.62.39	10.0.2.15	HTTP/X…	1337	HTTP/1.1 200 OK

DNS



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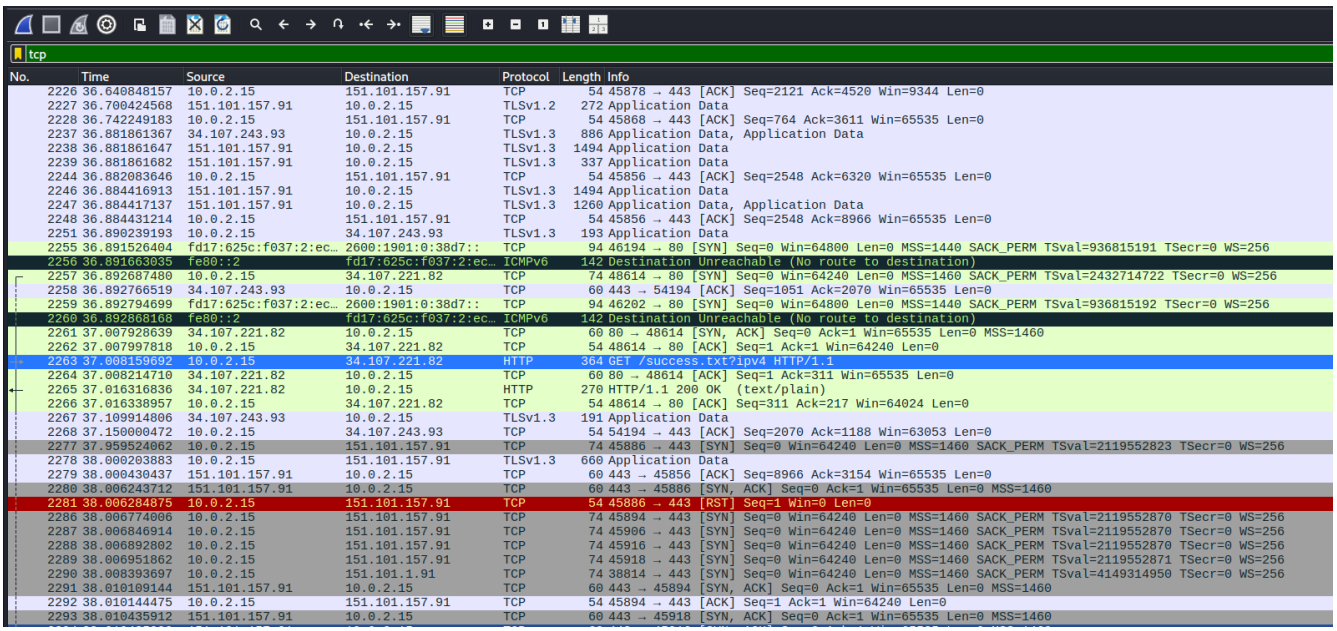
dns

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.0.2.15	172.20.72.11	DNS	80	Standard query 0x1cc8 A hareekshith.is-a.dev
2	0.000093557	10.0.2.15	172.20.72.11	DNS	70	Standard query 0x7c65 HTTPS github.com
3	0.000200784	10.0.2.15	172.20.72.11	DNS	70	Standard query 0xfcb4 A github.com
4	0.000216938	10.0.2.15	172.20.72.11	DNS	70	Standard query 0x34b5 AAAA github.com
5	0.000305102	10.0.2.15	172.20.72.11	DNS	80	Standard query 0x4d1a A hareekshith.is-a.dev
6	0.000317854	10.0.2.15	172.20.72.11	DNS	80	Standard query 0x501d AAAA hareekshith.is-a.dev
7	0.006397486	172.20.72.11	10.0.2.15	DNS	96	Standard query response 0x4d1a A hareekshith.is-a.dev A 216.198.79.1
8	0.006397751	172.20.72.11	10.0.2.15	DNS	154	Standard query response 0x34b5 AAAA github.com SOA ns-1707.awsdns-21.co.uk
9	0.006397819	172.20.72.11	10.0.2.15	DNS	154	Standard query response 0x7c65 HTTPS github.com SOA ns-1707.awsdns-21.co.uk
10	0.006397862	172.20.72.11	10.0.2.15	DNS	96	Standard query response 0x1cc8 A hareekshith.is-a.dev A 216.198.79.1
11	0.006397906	172.20.72.11	10.0.2.15	DNS	132	Standard query response 0x501d AAAA hareekshith.is-a.dev SOA ns1.is-a.dev
18	0.039759578	172.20.72.11	10.0.2.15	DNS	86	Standard query response 0xfcb4 A github.com A 20.207.73.82
240	1.906671984	10.0.2.15	172.20.72.11	DNS	71	Standard query 0x8233 A chatgpt.com
241	1.906703163	10.0.2.15	172.20.72.11	DNS	71	Standard query 0x0f3f AAAA chatgpt.com
242	1.909987204	172.20.72.11	10.0.2.15	DNS	103	Standard query response 0x8233 A chatgpt.com A 172.64.155.209 A 104.18.32.47
243	1.909987650	172.20.72.11	10.0.2.15	DNS	127	Standard query response 0x0f3f AAAA chatgpt.com AAAA 2a06:98c1:310b::ac40:9bd1 AAAA 2a06:98c1:3100::6
1043	2.339708135	10.0.2.15	172.20.72.11	DNS	84	Standard query 0x3a2a HTTPS persistent.oaistatic.com
1045	2.340044820	10.0.2.15	172.20.72.11	DNS	84	Standard query 0x48dc A persistent.oaistatic.com
1046	2.340056722	10.0.2.15	172.20.72.11	DNS	84	Standard query 0xecdd AAAA persistent.oaistatic.com
1208	2.431782918	172.20.72.11	10.0.2.15	DNS	116	Standard query response 0x48dc A persistent.oaistatic.com A 104.18.41.158 A 172.64.146.98
1209	2.431783013	172.20.72.11	10.0.2.15	DNS	146	Standard query response 0x3a2a HTTPS persistent.oaistatic.com SOA desiree.ns.cloudflare.com
1211	2.431783107	172.20.72.11	10.0.2.15	DNS	140	Standard query response 0xecdd AAAA persistent.oaistatic.com AAAA 2606:4700:440c::6812:299e AAAA 2a06
1304	3.624205717	10.0.2.15	172.20.72.11	DNS	74	Standard query 0xea75 A ab.chatgpt.com
1305	3.624235378	10.0.2.15	172.20.72.11	DNS	74	Standard query 0x867a AAAA ab.chatgpt.com
1306	3.626220541	172.20.72.11	10.0.2.15	DNS	106	Standard query response 0xea75 A ab.chatgpt.com A 172.64.155.209 A 104.18.32.47
1307	3.626220918	172.20.72.11	10.0.2.15	DNS	130	Standard query response 0x867a AAAA ab.chatgpt.com AAAA 2a06:98c1:3100::6812:202f AAAA 2a06:98c1:310b
2062	15.426489002	10.0.2.15	172.20.72.11	DNS	80	Standard query 0x1608 A hareekshith.is-a.dev
2063	15.426532630	10.0.2.15	172.20.72.11	DNS	80	Standard query 0xf615 AAAA hareekshith.is-a.dev
2064	15.430917132	172.20.72.11	10.0.2.15	DNS	96	Standard query response 0x1608 A hareekshith.is-a.dev A 216.198.79.1

TCP



tcp

No.	Time	Source	Destination	Protocol	Length	Info
2226	36.640848157	10.0.2.15	151.101.157.91	TCP	54	45878 → 443 [ACK] Seq=2121 Ack=4520 Win=9344 Len=0
2227	36.700424568	151.101.157.91	10.0.2.15	TLSv1.2	272	Application Data
2228	36.742249183	10.0.2.15	151.101.157.91	TCP	54	45868 → 443 [ACK] Seq=764 Ack=3611 Win=65535 Len=0
2237	36.881861367	34.107.243.93	10.0.2.15	TLSv1.3	886	Application Data, Application Data
2238	36.881861647	151.101.157.91	10.0.2.15	TLSv1.3	1494	Application Data
2239	36.881861682	151.101.157.91	10.0.2.15	TLSv1.3	337	Application Data
2244	36.882083646	10.0.2.15	151.101.157.91	TCP	54	45856 → 443 [ACK] Seq=2548 Ack=6320 Win=65535 Len=0
2246	36.884416913	151.101.157.91	10.0.2.15	TLSv1.3	1494	Application Data
2247	36.884417137	151.101.157.91	10.0.2.15	TLSv1.3	1260	Application Data, Application Data
2248	36.884431214	10.0.2.15	151.101.157.91	TCP	54	45856 → 443 [ACK] Seq=2548 Ack=8966 Win=65535 Len=0
2251	36.890239193	10.0.2.15	34.107.243.93	TLSv1.3	193	Application Data
2255	36.891526404	fd17:625c:f037:2:ec…	2600:1901:0:38d7::	TCP	94	46194 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=936815191 TSecr=0 WS=256
2256	36.891663035	fe80::2	fd17:625c:f037:2:ec…	ICMPv6	142	Destination Unreachable (No route to destination)
2257	36.892687480	10.0.2.15	34.107.221.82	TCP	74	48614 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2432714722 TSecr=0 WS=256
2258	36.892766519	34.107.243.93	10.0.2.15	TCP	60	443 → 54194 [ACK] Seq=1051 Ack=2070 Win=65535 Len=0
2259	36.892794699	fd17:625c:f037:2:ec…	2600:1901:0:38d7::	TCP	94	46202 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=936815192 TSecr=0 WS=256
2260	36.892868168	fe80::2	fd17:625c:f037:2:ec…	ICMPv6	142	Destination Unreachable (No route to destination)
2261	37.007928639	34.107.221.82	10.0.2.15	TCP	60	80 → 48614 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
2262	37.007997818	10.0.2.15	34.107.221.82	TCP	54	48614 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
2263	37.008159692	10.0.2.15	34.107.221.82	HTTP	364	GET /success.txt?ipv4 HTTP/1.1
2264	37.008214710	34.107.221.82	10.0.2.15	TCP	60	80 → 48614 [ACK] Seq=1 Ack=311 Win=65535 Len=0
2265	37.016316836	34.107.221.82	10.0.2.15	HTTP	270	HTTP/1.1 200 OK (text/plain)
2266	37.016338957	10.0.2.15	34.107.221.82	TCP	54	48614 → 80 [ACK] Seq=311 Ack=217 Win=64024 Len=0
2267	37.109914806	34.107.243.93	10.0.2.15	TLSv1.3	191	Application Data
2268	37.150000472	10.0.2.15	34.107.243.93	TCP	54	54194 → 443 [ACK] Seq=2070 Ack=1188 Win=63053 Len=0
2277	37.959524062	10.0.2.15	151.101.157.91	TCP	74	45886 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2119552823 TSecr=0 WS=256
2278	38.000203883	10.0.2.15	151.101.157.91	TLSv1.3	660	Application Data
2279	38.000430437	151.101.157.91	10.0.2.15	TCP	60	443 → 45856 [ACK] Seq=8966 Ack=3154 Win=65535 Len=0
2280	38.006243712	151.101.157.91	10.0.2.15	TCP	60	443 → 45886 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
2281	38.006284875	10.0.2.15	151.101.157.91	TCP	54	45886 → 443 [RST] Seq=1 Win=0 Len=0
2286	38.006774006	10.0.2.15	151.101.157.91	TCP	74	45894 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2119552870 TSecr=0 WS=256
2287	38.006846914	10.0.2.15	151.101.157.91	TCP	74	45906 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2119552870 TSecr=0 WS=256
2288	38.006892802	10.0.2.15	151.101.157.91	TCP	74	45916 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2119552870 TSecr=0 WS=256
2289	38.006951862	10.0.2.15	151.101.157.91	TCP	74	45918 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2119552871 TSecr=0 WS=256
2290	38.008393697	10.0.2.15	151.101.1.91	TCP	74	38814 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=4149314950 TSecr=0 WS=256
2291	38.010109144	151.101.157.91	10.0.2.15	TCP	60	443 → 45894 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
2292	38.010144475	10.0.2.15	151.101.157.91	TCP	54	45894 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
2293	38.010435912	151.101.157.91	10.0.2.15	TCP	60	443 → 45918 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460

Note: There are also 3 other protocols such as QUIC (Quick UDP internet Connections), TLSv1.3 (Transport Layer Security), ICMP(Internet Control Message Protocol) and OSCP (Online Security Certification Protocol).

Step 4: Export the whole capture into a .pcap file for future references.

- Hareekshith AS